

04 Determinants



4.1 The 2×2 Determinant

Understanding the 2×2 matrix determinant through parallelogram area

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills for This Section

1

How the determinant behaves: defined only for square matrices, and always a scalar

2

The 2×2 determinant represents the signed area of a parallelogram

3

The identity matrix corresponds to the unit square

4

A diagonal matrix with equal diagonal entries: columns form a square

5

A diagonal matrix with unequal diagonal entries: columns form a rectangle

6

An upper-triangular matrix: columns form a parallelogram sheared along the horizontal axis

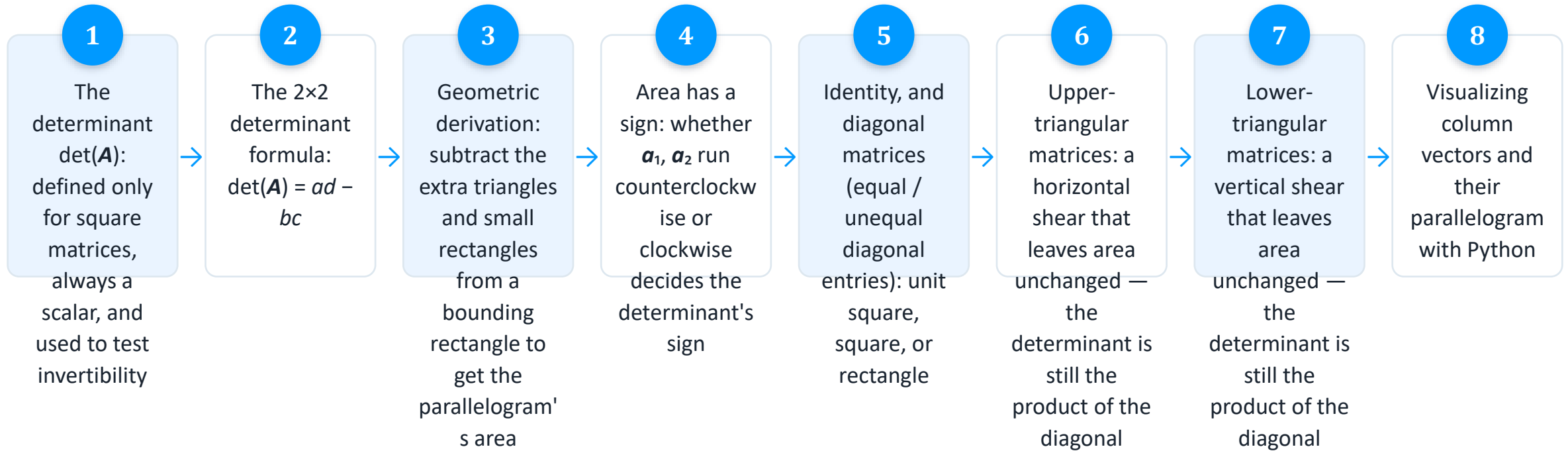
7

A lower-triangular matrix: columns form a parallelogram sheared along the vertical axis

8

A square matrix with determinant 0 is not invertible (covered in depth in the next chapter)

How This Section Unfolds



The Determinant of a Square Matrix Is a Scalar

The determinant, $\det()$ — only square matrices have one, and the result is always a single scalar

- The determinant tells us whether a matrix is invertible: a square matrix is invertible exactly when its determinant is nonzero; a zero determinant makes it a singular matrix, not invertible
- Geometrically, the determinant is closely tied to how area and volume change: a linear transformation's determinant is its scaling factor for area or volume
- Determinants are also used to test linear dependence among vectors and to compute eigenvalues — concepts essential to linear transformations, differential equations, computer graphics, and machine learning

Given a 2×2 Matrix A , the Determinant Is $ad - bc$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \det(A) = ad - bc$$

- This book uses $\det()$ as the determinant operator — $\det$ comes from the first three letters of “determinant”
- The determinant only applies to square matrices: a square matrix has equal row and column counts, an $n \times n$ matrix; non-square matrices have no determinant
- The determinant operation maps a square matrix to a scalar: however many entries the matrix has, the result is always a single number

Understanding the Determinant Through Parallelogram Area

Split A into two column vectors $\mathbf{a}_1 = [a;c]$, $\mathbf{a}_2 = [b;d]$, and plot them in the plane, both starting at the origin

- The parallelogram formed by $\mathbf{a}_1$ and $\mathbf{a}_2$ is exactly the one from the parallelogram rule for vector addition
- The parallelogram's four vertices are: the origin (0, 0); the tip of $\mathbf{a}_1$ at (a, c); the tip of $\mathbf{a}_2$ at (b, d); and the tip of their sum at (a+b, c+d)

Figure 1 · Constructing the Parallelogram

- The red arrow is $\mathbf{a}_1 = [a;c]$, the green arrow is $\mathbf{a}_2 = [b;d]$, and the orange diagonal is their sum $\mathbf{a}_1 + \mathbf{a}_2 = [a+b;c+d]$
- The right-hand panel marks each side's projection onto the horizontal and vertical axes — lengths a, b, c, d , the key to building the bounding rectangle and deriving

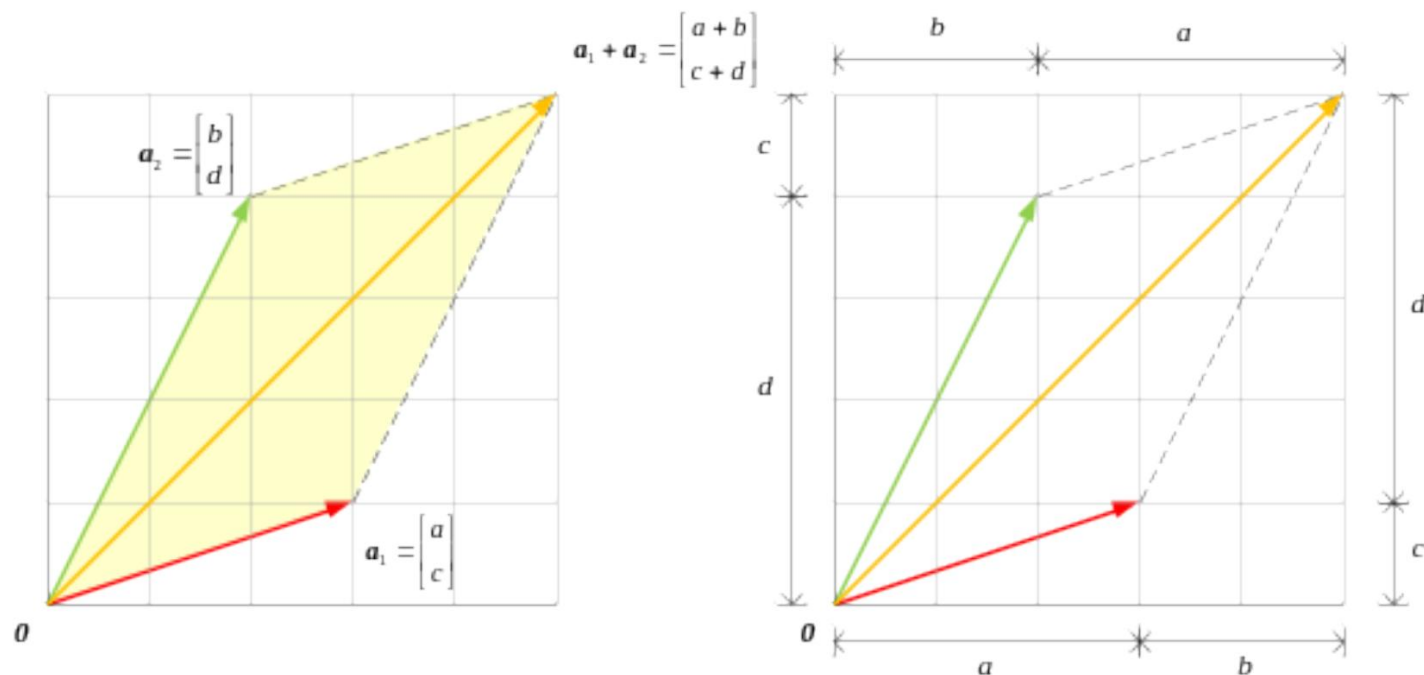


图 1. 构造平行四边形

Build a Rectangle First, Then Subtract the Extra Parts

The parallelogram is contained in a bounding rectangle whose diagonal corners are $(0,0)$ and $(a+b, c+d)$

- This rectangle contains parts that don't belong to the parallelogram: four right triangles (shown in pink and green) and two small rectangles (shown in gray)
- The parallelogram's area can be found by subtracting these extra parts from the bounding rectangle's area

Figure 2 · Building and Slicing the Rectangle

- The bounding rectangle has width $a+b$ and height $c+d$; the yellow region is the parallelogram we want
- The pink and green triangles and the gray small rectangles don't belong to the parallelogram — they are what we subtract next

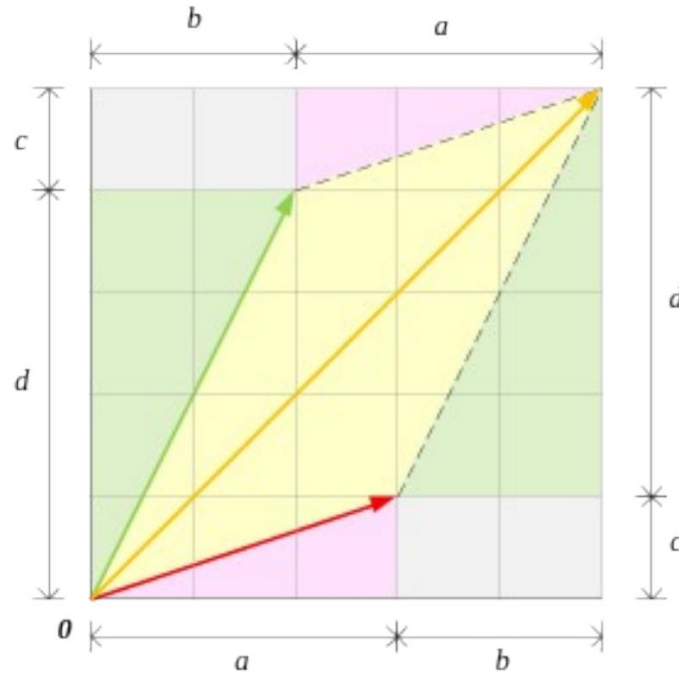


图 2. 构造并切割矩形

Bounding-Rectangle Area – Extra Parts = $ad - bc$

$$\det(A) = ad - bc$$

- Subtracting the four triangles and two small rectangles from the bounding rectangle's area gives exactly the determinant of the 2×2 matrix $\mathbf{A}$ — the geometric meaning of the 2×2 determinant
- Note: in this section a parallelogram's area can be positive or negative; in the next section, so can a parallelepiped's volume

Figure 3 · Computing the Parallelogram's Area

- Yellow parallelogram = blue bounding rectangle – (pink triangles + green triangles + gray small rectangles)
- The result of this subtraction is exactly $\det(\mathbf{A}) = ad - bc$

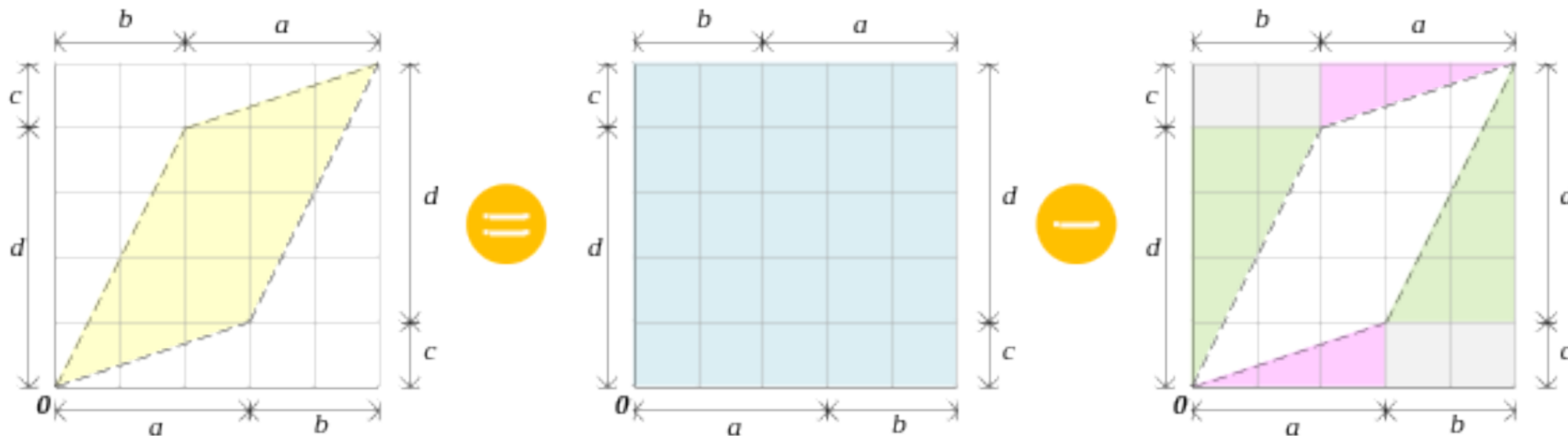


图 3. 计算平行四边形面积

The Identity Matrix Has Determinant 1

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \det(A) = 1 \times 1 - 0 \times 0 = 1$$

- $\mathbf{a}_1 = [1; 0]$ and $\mathbf{a}_2 = [0; 1]$ form a unit square, with area exactly 1
- By this section's convention, red is the first column vector $\mathbf{a}_1$ and green is the second column vector $\mathbf{a}_2$; going counterclockwise (within 180°), red comes before green, and the determinant is positive

Figure 4 · Unit Squares, Set One

- (a) $\mathbf{a}_1$ before $\mathbf{a}_2$, going counterclockwise, so the determinant is positive: $\det([1,0;0,1]) = 1$
- (b) After swapping the columns, $\mathbf{a}_1 = [0;1]$, $\mathbf{a}_2 = [1;0]$: the order becomes clockwise, equivalent to flipping the coordinate system once, and the determinant flips sign: $\det([0,1;1,0]) = -1$

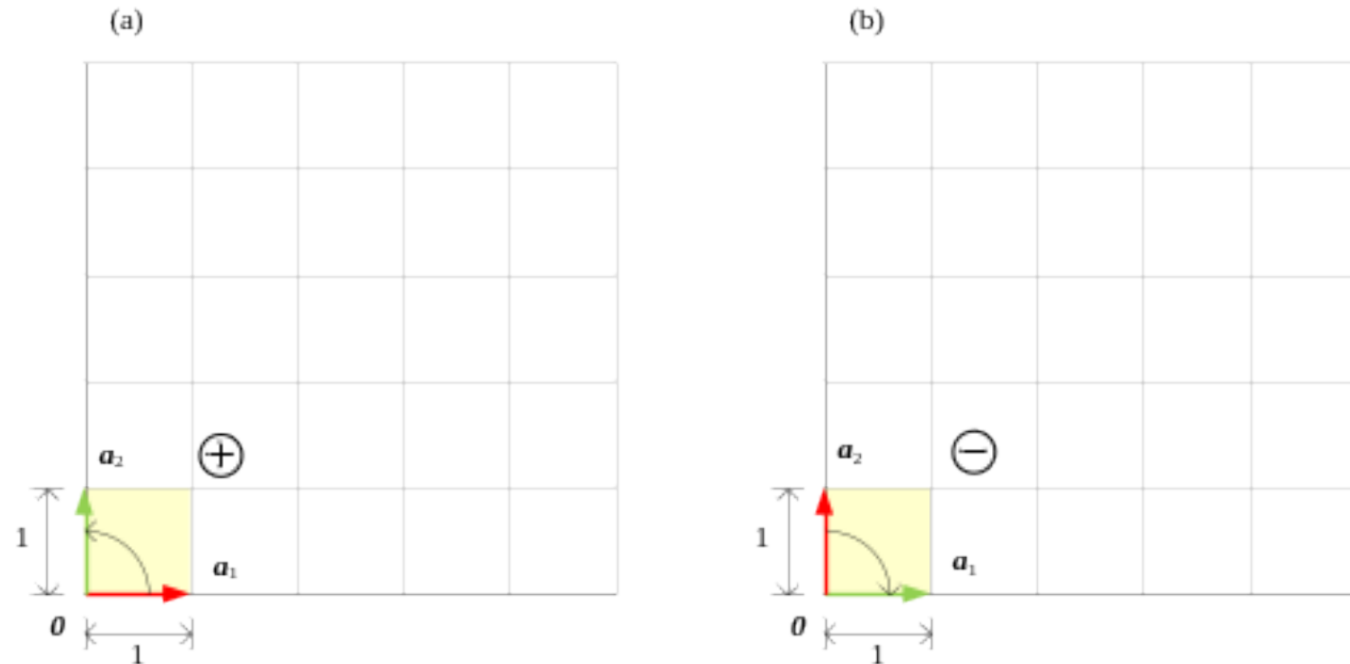


图 4. 单位正方形，第一组

Figure 5 · Unit Squares, Set Two

- (a) These two vectors still form a unit square, but going counterclockwise, green $\mathbf{a}_2$ comes before red $\mathbf{a}_1$, so the determinant is negative: $\det\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -1$
- (b) Analyze this 2×2 matrix: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (this is the identity matrix, which corresponds to no flip)

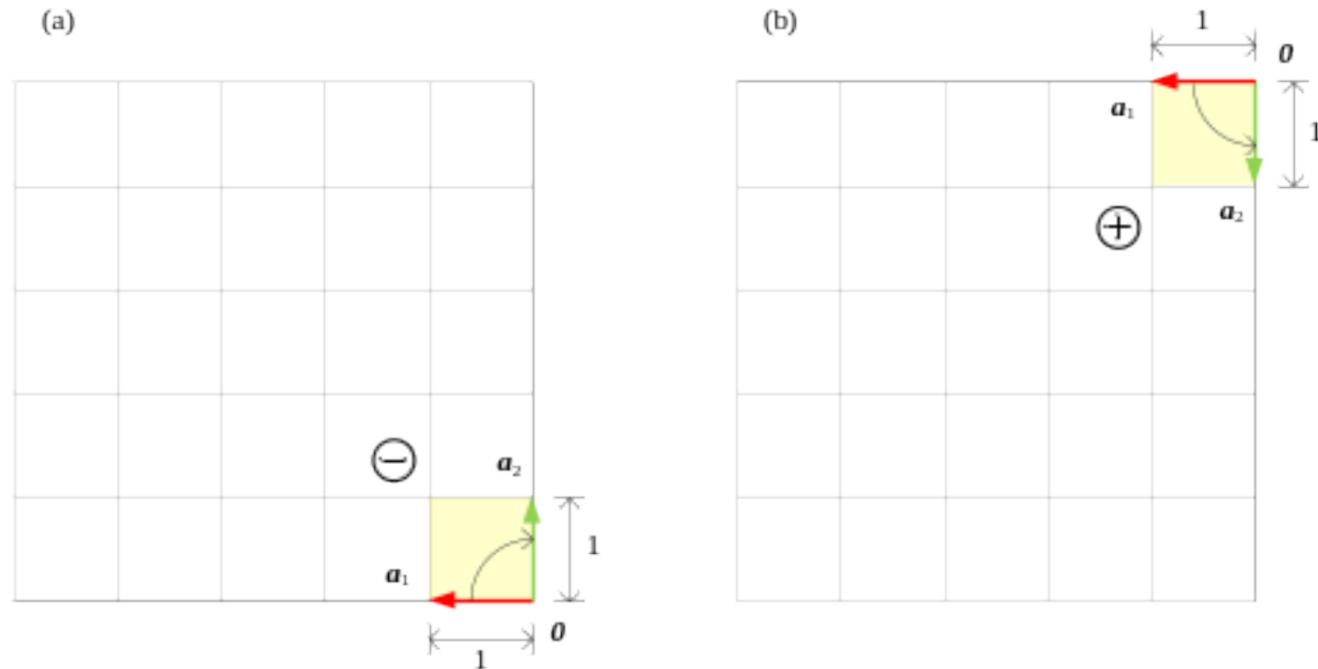


图 5. 单位正方形，第二组

Equal Diagonal Entries: a Square

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \Rightarrow \det(A) = 3 \times 3 - 0 \times 0 = 9$$

- $\mathbf{a}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$ form a square with side length 3, area 9 — exactly the determinant's value
- A diagonal matrix acts like scaling a shape uniformly along the horizontal and vertical directions

Figure 6 · A Square

- (a) $\mathbf{a}_1$ and $\mathbf{a}_2$ point in different directions but have the same length (both 3), forming a square, with positive determinant
- (b) Analyze this 2x2 matrix's determinant on your own

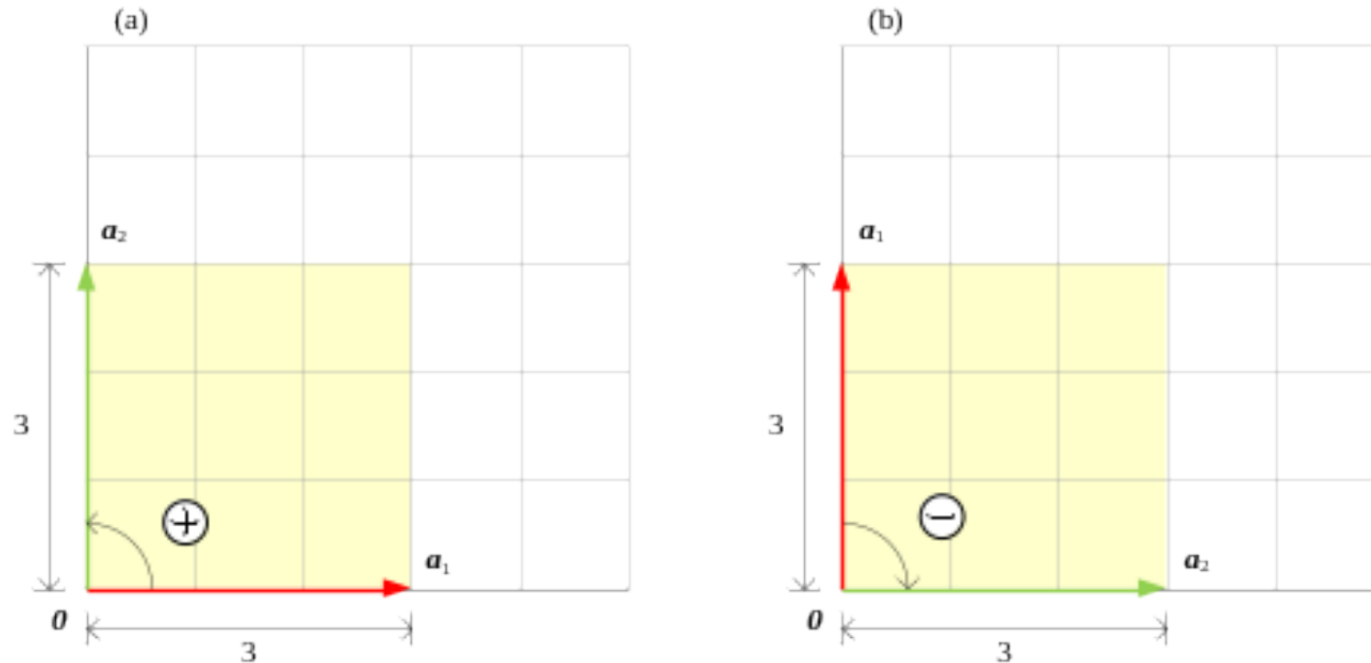


图 6. 正方形

Unequal Diagonal Entries: a Rectangle

$$\mathbf{A} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow \det(\mathbf{A}) = 4 \times 2 = 8$$

- When 2×2 matrix $\mathbf{A}$ is diagonal with unequal main-diagonal entries, its geometric shape is a rectangle, with area “length × width”
- A diagonal matrix scales a shape by different factors along the horizontal and vertical directions

Figure 7 · A Rectangle

- (a) $\mathbf{a}_1 = [4;0]$, $\mathbf{a}_2 = [0;2]$ form a 4×2 rectangle, with area 8
- (b) After swapping the two columns, $\mathbf{a}_1 = [0;2]$, $\mathbf{a}_2 = [4;0]$: the rectangle's shape is unchanged, but watch the determinant's sign here

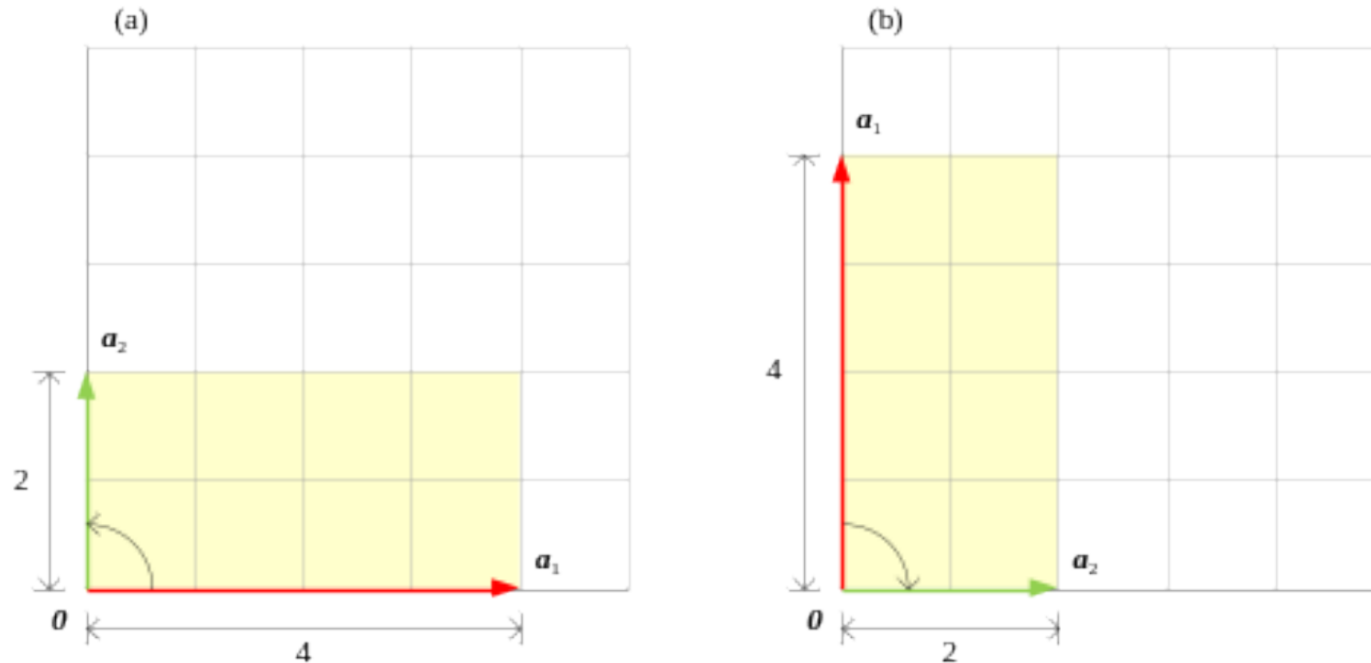


图 7. 矩形

A Diagonal Matrix's Determinant Is the Product of Its Diagonal

$$\text{diagonal matrix } \text{diag}(d_1, d_2, \dots, d_n) \Rightarrow \det(A) = d_1 \times d_2 \times \dots \times d_n$$

- The examples above show that a diagonal matrix's determinant equals the product of its main-diagonal entries
- This also tells us: if any main-diagonal entry of a diagonal matrix is 0, that matrix's determinant is 0

Upper-Triangular: Horizontal Shear, Area Unchanged

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \Rightarrow \det(A) = 3 \times 2 = 6$$

- The first column vector $\mathbf{a}_1$ always points parallel to the x1 axis; its x1-component is exactly the parallelogram's base length
- Once $\mathbf{a}_2$'s x2-component (the parallelogram's height) is fixed, no matter how $\mathbf{a}_2$'s x1-component changes, the parallelogram's area stays the same — that component only changes how much the shape leans

Figure 8 · Upper-Triangular Matrices

- (a) $\det([3,1;0,2]) = 6$; (b) $\det([3,2;0,2]) = 6$ — the two upper-triangular matrices clearly share the same determinant
- Geometrically, this transformation applies a horizontal shear (plus overall scaling) to the parallelogram: stretching or compressing along the x_1 axis without changing the height

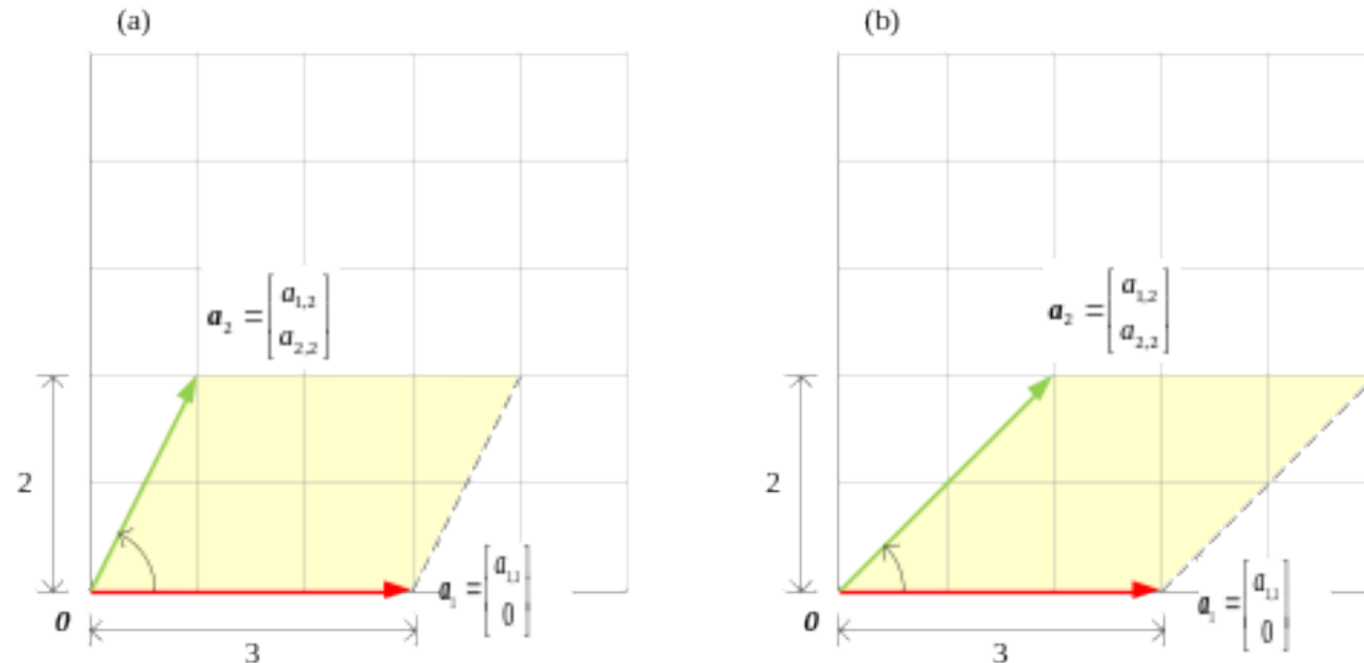


图 8. 上三角矩阵

Lower-Triangular: Vertical Shear, Area Unchanged

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \Rightarrow \det(A) = 2 \times 3 = 6$$

- With upper-triangular matrices understood, lower-triangular ones are easy: the second column vector $\mathbf{a}_2$ always points parallel to the x2 axis
- The first column vector $\mathbf{a}_1$'s x1-component determines the parallelogram's height; the area has nothing to do with $\mathbf{a}_1$'s x2-component

Figure 9 · Lower-Triangular Matrices

- (a) $\det([2,0;1,3]) = 6$; (b) $\det([2,0;2,3]) = 6$ — the two lower-triangular matrices likewise share the same determinant
- Geometrically, this transformation applies a vertical shear (plus scaling) to the parallelogram: the base length and vertical height directly determine the final area

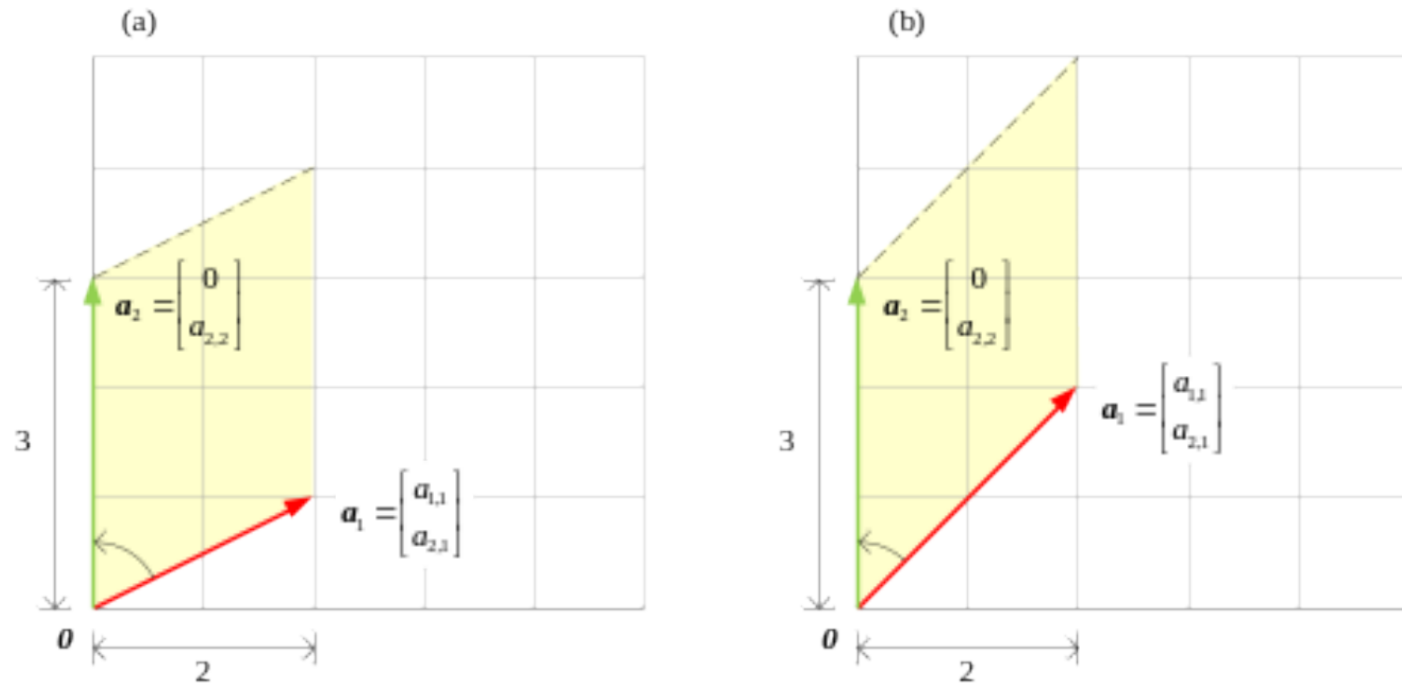


图 9. 下三角矩阵

A Triangular Matrix's Determinant Is Also the Product of Its Diagonal

upper- (or lower-) triangular, with diagonal entries $d_1, d_2, \dots, d_n \implies \det(A) = d_1 \times d_2 \times \dots \times d_n$

- An upper-triangular matrix's determinant is decided purely by its main-diagonal entries, since they correspond to the parallelogram's base length and vertical height, which together determine the final area
- Like diagonal and upper-triangular matrices, a lower-triangular matrix's determinant also equals the product of its main-diagonal entries; if any diagonal entry is 0, the determinant is 0

Visualizing a Matrix's Columns and Parallelogram in Python

First extract the matrix's two column vectors, build the parallelogram's four vertices, then draw the vectors and fill the region with `arrow()` and `fill()`

- $A[:, 0]$ and $A[:, 1]$ extract $\mathbf{A}$'s first and second columns; the four vertices $P1=\text{origin}$, $P2=a1$, $P3=a1+a2$, $P4=a2$ form the parallelogram
- `ax.arrow()` draws red vector $a1$ and green vector $a2$; `ax.fill()` fills the parallelogram with a dashed black border, 30% opacity, and yellow fill

VISUALIZING A 2×2 MATRIX

Code 1 · Visualizing the Parallelogram of a 2×2 Matrix

LA_04_01_01.ipynb

- a: create a 2D array $A = \text{np.array}([[2,1],[1,3]])$ representing matrix A
- b: $A[:, 0]$ extracts column vector a_1 , $A[:, 1]$ extracts column vector a_2
- c: build the four vertices P_1 – P_4 of the parallelogram, packed into the point sequence `parallelogram` with `np.array()`
- d: draw vectors a_1 (red) and a_2 (green) with `ax.arrow()`, both starting at the origin
- e: draw the parallelogram from the vertex sequence with `ax.fill()` — black dashed border, 30% opacity, yellow fill

```
## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 定义一个 2x2 的矩阵 A
A = np.array([[2, 1],
              [1, 3]])

## 提取两列向量`
a1 = A[:, 0] # 第一列向量
a2 = A[:, 1] # 第二列向量

## 构建平行四边形
origin = np.array([0, 0])
# 起点为原点
P1 = origin
P2 = a1
P3 = a1 + a2
P4 = a2
parallelogram = np.array([P1, P2, P3, P4])

## 可视化
fig, ax = plt.subplots()

# 绘制向量箭头
ax.arrow(*origin, *a1, head_width=0.2,
        length_includes_head=True, color='red')
ax.arrow(*origin, *a2, head_width=0.2,
        length_includes_head=True, color='green')

# 绘制平行四边形
ax.fill(parallelogram[:, 0], parallelogram[:, 1],
       edgecolor='k', linestyle='--',
       alpha=0.3, facecolor='yellow')

# 图像装饰
ax.set_aspect('equal')
ax.set_xlim(-1, max(P3[0], P2[0], P4[0]) + 1)
ax.set_ylim(-1, max(P3[1], P2[1], P4[1]) + 1)
ax.axhline(0, color='k', lw=1)
ax.axvline(0, color='k', lw=1)
ax.grid(True, color='0.88')
```

The Determinant: a Square Matrix's “Signed Area / Volume” Ruler

$\det(A) = ad - bc$ — the 2×2 determinant is the signed area of the parallelogram formed by the column vectors

- Identity $\rightarrow$ unit square; diagonal $\rightarrow$ square or rectangle; upper-/lower-triangular $\rightarrow$ a sheared parallelogram with unchanged area — all of these have determinants equal to the product of their diagonal entries
- The determinant's sign is set by the column vectors' arrangement (counterclockwise is positive, clockwise is negative); a determinant of 0 means the matrix is not invertible, which the next chapter covers in depth

Key Terms in This Section

Determinant $\det(A)$ — defined only for square matrices, always evaluating to a scalar

Invertible / Singular $\det(A) \neq 0$ means A is invertible; $\det(A) = 0$ makes A singular and non-invertible

Unit Square The square formed by the identity matrix's two column vectors, with area 1

Upper-Triangular Matrix All entries below the main diagonal are 0; geometrically a horizontal shear that preserves area

Determinant of a Triangular Matrix Also equals the product of the main-diagonal entries, independent of the off-diagonal entries

2x2 Determinant Formula $\det(A) = ad - bc$, where $A = [a,b;c,d]$

Signed Area The determinant's absolute value equals the parallelogram's area; the sign follows the column vectors' arrangement

Determinant of a Diagonal Matrix Equals the product of the main-diagonal entries: $d_1 \times d_2 \times \cdots \times d_n$

Lower-Triangular Matrix All entries above the main diagonal are 0; geometrically a vertical shear that preserves area

Singular Matrix A square matrix with determinant 0 — not invertible

GO FURTHER

Companion Code and Further Reading

Further Reading · LA_04_01_01.ipynb

Visualize the parallelogram of a 2×2 matrix; practice extracting column vectors and plotting

Further Reading · Determinants and Invertibility

A later chapter is devoted to inverse matrices; $\det(A) \neq 0$ is the necessary and sufficient condition for A to be invertible

Coming Up · Next Section

The determinant of a 3×3 matrix: from parallelogram area to parallelepiped volume

Open-Source Repository

Code and figures for the whole book: <https://github.com/Visualize-ML>

Questions Q1 – Q9

Q1 Compute the determinant of each diagonal matrix below, and sketch the parallelogram formed by its column vectors on grid paper: ① $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$; ② $\begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$; ③ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$; ④ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$; ⑤ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

Q3 Compute the determinant of each lower-triangular matrix below, and sketch the corresponding parallelogram: ① $\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$; ② $\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 8 \end{bmatrix}$; ③ $\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 8 \end{bmatrix}$; ④ $\begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 8 \end{bmatrix}$; ⑤ $\begin{bmatrix} 1 & 0 & 8 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 8 \end{bmatrix}$.

Q5 Explore the connection between the determinant and the cross product of vectors.

Q7 Write a function `parallelogram_area(A)` that returns the area of the parallelogram formed by A's two column vectors (the absolute value of the determinant), and plot the parallelogram with `matplotlib`.

Q9 Write a function `sign_of_det(A)` that returns the sign of A's determinant (positive, negative, or zero), and explain its geometric meaning in terms of the column vectors' direction.

Q2 Compute the determinant of each upper-triangular matrix below, and sketch the corresponding parallelogram: ① $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 2 & 8 & 0 & 0 \end{bmatrix}$; ② $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 2 & 8 & 0 & 0 \end{bmatrix}$; ③ $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 2 & 8 & 0 & 0 \end{bmatrix}$; ④ $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 2 & 8 & 0 & 0 \end{bmatrix}$; ⑤ $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 2 & 2 & 0 & 3 \\ 0 & 0 & 0 & 0 & 2 & 8 & 0 & 0 \end{bmatrix}$.

Q4 Swap the column vectors in the matrices from Q1, Q2, and Q3 (i.e. swap the matrix's two columns), recompute the determinants, and compare with the originals.

Q6 Write a Python function `det_2x2(A)` that manually implements the ad – bc formula for a 2×2 matrix, and compare it against `np.linalg.det(A)`.

Q8 Construct 5 random 2×2 matrices and check whether `det(A)` always equals `det(A transposed)`; explain why using the parallelogram picture.

Questions Q10 – Q18

Q10 Write a function `swap_columns_det(A)` that swaps A's two columns and computes the determinant, verifying it is always the negative of the original.

Q12 Write a function `scale_column_det(A, k, col)` that multiplies A's given column by scalar k and computes the determinant; explore how the determinant changes with k, and state the pattern you find.

Q14 Write a function `add_column_multiple(A, k)` that adds k times A's first column to its second column, verify the determinant stays unchanged, and explain why.

Q16 Write a function `triangular_det(A)` that checks whether A is upper- or lower-triangular, returning the product of the diagonal directly if so, and calling `np.linalg.det()` otherwise.

Q18 Write a function `visualize_det_sign(A)` that fills the parallelogram formed by the column vectors in different colors depending on the determinant's sign (e.g. blue for positive, red for negative).

Q11 Construct a 2×2 matrix with linearly dependent columns (e.g. $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$), compute its determinant, and explain why it's zero in terms of the parallelogram “collapsing” into a line segment.

Q13 With a Python animation (or a series of subplots), fix $a_1 = [1, 0]$ and rotate a_2 continuously from $[0, 1]$ to $[1, 0]$; observe how the parallelogram's area (the determinant) changes continuously down to 0.

Q15 Construct a 2×2 permutation matrix (swap the identity matrix's two columns), compute its determinant, and cross-check it against your findings in Q4.

Q17 Construct 10 random 2×2 matrices and compare your `det_2x2` function against `np.linalg.det()` numerically, discussing any discrepancy that floating-point computation might introduce.

Questions Q19 – Q25

Q19 Explore: given the identity matrix I , construct kI for scalar k , compute its determinant, form a conjecture about the relationship between $\det(kI)$, k , and the matrix's dimension, and verify it in Python.

Q21 Building on this section, write a function `shear_matrix(k, direction)` that generates a horizontal or vertical shear matrix (upper- or lower-triangular with a diagonal of 1s), and verify its determinant is always 1.

Q23 Write a function `random_singular_matrix()` that randomly generates a singular (determinant-zero) 2×2 matrix, and plot it to show why its column vectors are collinear.

Q25 Building on this section, write a short summary (comments are fine) covering how the determinant is computed for the identity, diagonal, upper-triangular, and lower-triangular matrices, and what each means geometrically.

Q20 Write a function `is_invertible(A)` that returns whether matrix A is invertible by checking whether its determinant is zero (use a tolerance such as $1e-9$ to avoid floating-point error).

Q22 Explore the connection between the determinant and the cross product (echoing Q5): verify in Python whether the “z-component of the cross product” of 2D vectors a , b equals the determinant of the matrix with columns a and b .

Q24 Explore: construct one upper-triangular matrix and one lower-triangular matrix, verify whether their product is still triangular, and check whether its determinant still equals the product of the two original determinants.

10 Prompts to Explore Further (1/2)

- 1 Explain why the determinant is only defined for square matrices, and makes no sense for non-square ones.
- 2 Explain how the 2×2 determinant formula $\det(A) = ad - bc$ is derived from a parallelogram's area.
- 3 Explain why swapping a matrix's two columns flips the sign of the determinant, and how this relates to “flipping” the coordinate system.
- 4 Explain why the determinants of diagonal, upper-triangular, and lower-triangular matrices all equal the product of the main-diagonal entries.
- 5 Explain why the “horizontal shear” transformation corresponding to an upper-triangular matrix never changes a parallelogram's area.

10 Prompts to Explore Further (2/2)

6

Explain the geometric meaning of a determinant equal to 0: why does the parallelogram formed by the column vectors “collapse” into a line segment?

7

Explain the relationship between the determinant and whether a matrix is invertible — why is a singular matrix not invertible?

8

Explain the connection between the determinant and the cross product of 2D vectors.

9

Explain how thinking of the determinant as an “area-scaling factor” helps in understanding linear transformations.

10

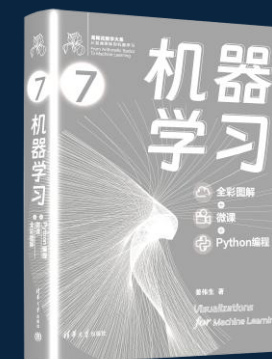
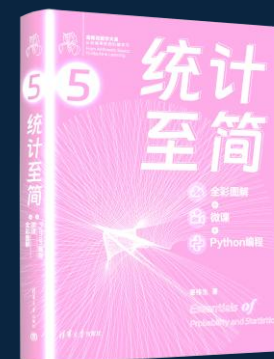
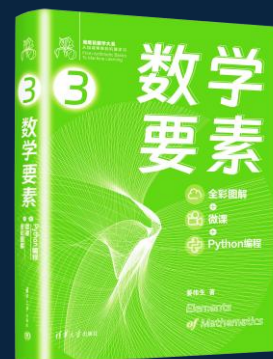
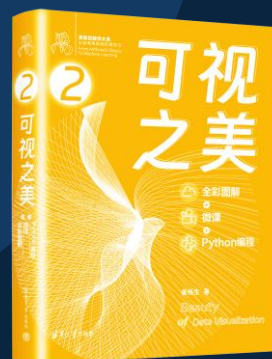
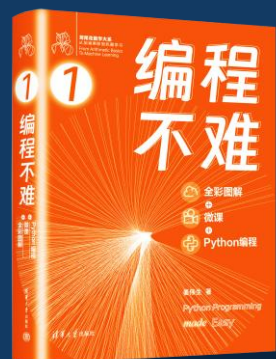
Building on this section, discuss why understanding the geometric meaning of the 2×2 determinant is an important foundation for learning the 3×3 determinant and determinants in higher dimensions.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

